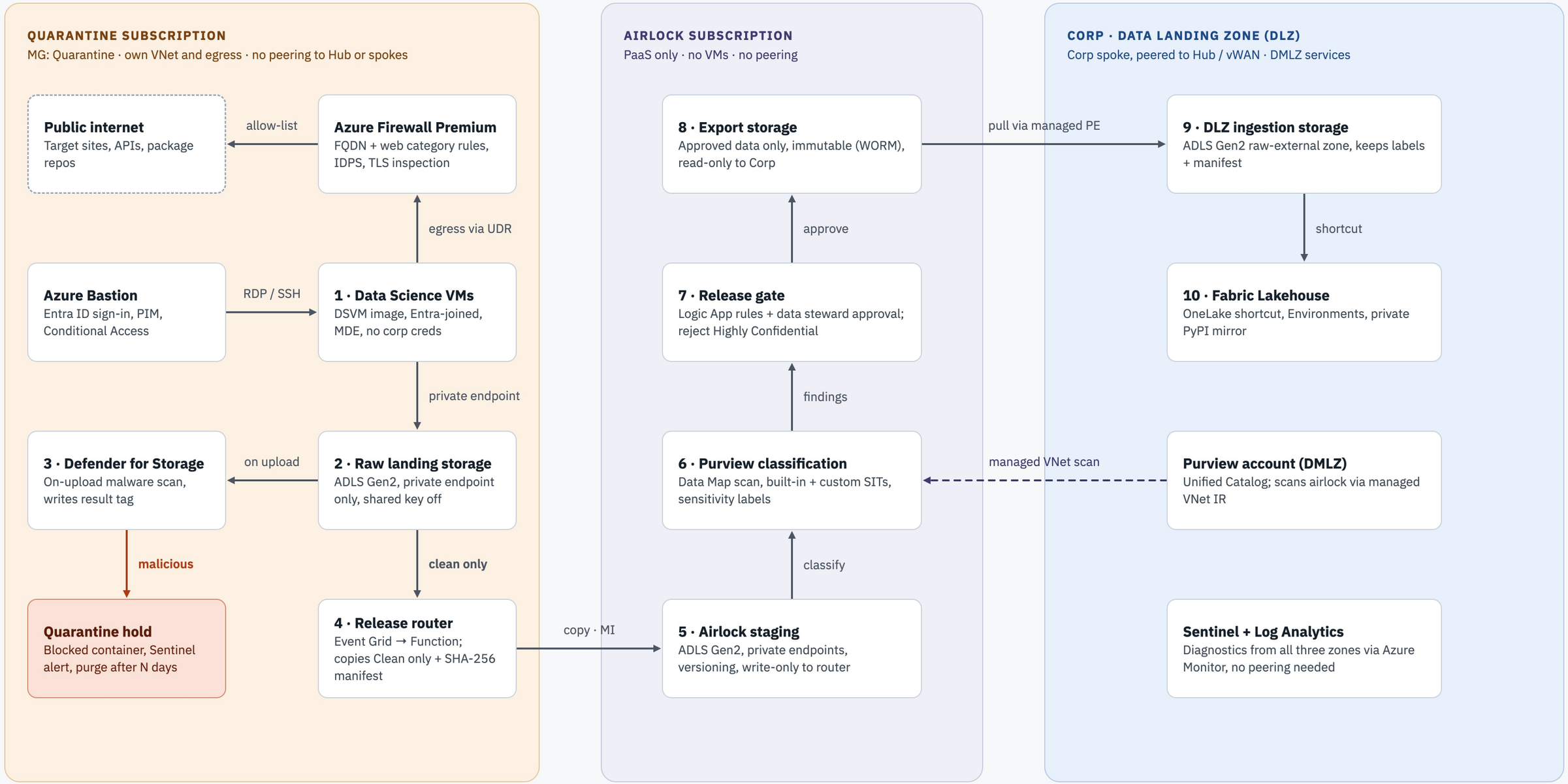
Internet data airlock on Azure

Scrape in isolation → land raw → malware scan → stage in airlock → classify → approve → release to Corp. One-way flow; the internet-facing zone has no network path to Hub or spokes. Dashed = management / scan path, not data.



Guardrails by control area

What has to be true in each zone for the airlock to hold. Tags show where each control lands: **Quarantine**, **Airlock**, **Corp**.

QUARANTINE · AIRLOCK

Network isolation

- Quarantine VNet has no peering, VPN or ExpressRoute gateway; Azure Policy denies `virtualNetworkPeerings` and gateways at the MG.
- All egress forced through Azure Firewall Premium (UDR 0.0.0.0/0); IDPS in deny mode, malware / phishing web categories blocked.
- Storage public network access disabled; private endpoints with private DNS zones local to the subscription, not linked to Hub DNS.
- Airlock hosts no VMs; Corp reaches export storage only through ADF / Purview managed private endpoints, approved on the airlock side.

ALL ZONES

Identity and access

- Dedicated Entra groups for quarantine users; VM login roles PIM-eligible, time-bound.
- Conditional Access on Bastion: MFA + compliant device; no clipboard / file copy-out (Bastion native client settings).
- DSVMs Entra-joined only, never AD DS joined; no corp secrets, Key Vault references or service connections on them.
- Router Function MI: read on raw, write-only on staging. Corp pipeline MI: read-only on export. Shared key and SAS disabled.

QUARANTINE

Malware and threat protection

- Defender for Servers P2 with MDE on DSVMs: AV, EDR, attack surface reduction rules.
- Defender for Storage on-upload malware scanning on raw; results to an Event Grid custom topic and blob index tag.
- Treat "unscannable" (oversize, encrypted or password-protected archives) as blocked, not clean. Check scan size and archive limits against real volumes.
- Optional: file-type allow-list and content disarm and reconstruction (CDR) for Office / PDF before release.

AIRLOCK

Classification and release

- Purview Data Map scans staging (scheduled, plus API-triggered per batch) with built-in AU SITs (TFN, Medicare) and custom classifiers.
- Sensitivity labels via Purview Information Protection auto-labelling where the file type supports it.
- Gate rules: no findings → auto-approve; PII → data steward; Highly Confidential → reject or route to a restricted DLZ container.
- Approval record (who, when, what, hashes) stored alongside the manifest.

PLATFORM

Governance (ALZ / Azure Policy)

- New "Quarantine" MG beside Corp / Online (not under Corp) with its own policy initiative; airlock sub under its own MG or Platform.
- Deny: peering, public IPs except firewall, storage public access, shared key. DINE: diagnostic settings, Defender plans.
- Budgets and auto-shutdown on DSVMs; ephemeral rebuild from a golden image if feasible.
- Bicep / AVM modules via Azure DevOps, one workload-identity service connection per subscription.

ALL ZONES → SOC

Monitoring and audit

- Diagnostic settings to central Log Analytics / Sentinel; travels over Azure Monitor, so no network path is needed.
- Analytics rules: malware found, peering or PE-approval attempts, unexpected egress volume, gate overrides.
- Storage read / write logs on raw, staging and export for full chain of custody.
- Immutable (WORM) retention on export; SHA-256 re-verified on arrival in the DLZ.

Design decisions and options

Five choices that shape the build. The highlighted option is the starting recommendation; the others stay on the table until the open questions are answered.

D1

Data scientist workspace

Internet scraping with DSVM-style tooling

RECOMMENDED

Data Science VM (Ubuntu / Windows)

Marketplace image with Python, R, VS Code, Jupyter preinstalled. Entra login, Bastion, auto-shutdown. Simple; you own patching.

Azure ML compute instances

Managed VNet with internet outbound allowed, idle shutdown. Workspace dependencies (Key Vault, ACR, storage) must also live in the quarantine sub.

AVD with DSVM tooling

Best control over clipboard and drive redirection, pooled cost. More moving parts: host pools, FSLogix, images.

D2

Quarantine → airlock transfer

Only malware-clean files move

RECOMMENDED

Event Grid + Function, server-side copy

Fires on the Defender scan-result event, copies Clean only with managed identity, writes a SHA-256 manifest. Per-file control and audit.

Storage object replication

No code, but filters by prefix only: needs a separate "clean" container first. Async, no hook for manifests or approvals.

ADF with managed VNet

Low-code with good monitoring. Adds another managed PE into quarantine storage and per-run cost; scan-tag filtering needs custom logic.

D3

Airlock → Corp release

Getting approved data into the DLZ / Fabric

RECOMMENDED

Corp pulls via managed PE

ADF or Fabric pipeline in the DLZ reads the read-only export account. Corp initiates; nothing in the airlock can write into Corp.

OneLake shortcut to export

Zero-copy via trusted workspace access. Data stays in airlock storage, so lifecycle, retention and ownership blur.

AVOID

Airlock pushes into DLZ

Simple, but opens an inbound write path into a Corp-connected storage account.

D4

Classification gate

Purview before ingestion

RECOMMENDED · SYSTEM OF RECORD

Purview Data Map scan + labels

Built-in and custom SITs, catalog and lineage carry into the DLZ. Scans are scheduled or API-triggered, so expect minutes-to-hours latency.

PAIR WITH PURVIEW IF SPEED MATTERS

Inline PII check in the Function

Azure AI Language PII detection or Presidio per file: event-driven, immediate verdict. Doesn't populate the catalog.

Defender CSPM sensitive data discovery

Posture view of sensitive data on storage using Purview classifiers. Useful signal, not a release gate on its own.

D5

Where the airlock lives

Ownership and blast radius

RECOMMENDED

Dedicated airlock subscription

Clean RBAC boundary neither side controls; operated by the platform or data governance team with its own policies.

Inside the quarantine subscription

Fewer subscriptions, but quarantine admins could tamper with the gate.

Inside the DLZ subscription

Close to consumers, but unclassified data sits inside a Corp-connected boundary.